

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0244

Jefferson Street 138 kV
January 01, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0244. The Interconnection Customer proposes a 322.1 MW natural gas combined cycle facility interconnecting at the Jefferson Street 138 kV substation in the Virginia area. The request was received on June 07, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 167 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,150,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DC WATER CHP – JEFFERSON STREET 138kV
Contingency: GUM SPRINGS – GRANTLINE-SIEFKER 138kV
Loading: 136.8% (Rating: 251 MVA)
DFAX: 60.7%
Pre-project loading: 59.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DC WATER CHP – JEFFERSON STREET 138kV	Rebuild line with higher-capacity conductor	\$1,150,000	100% (\$1,150,000)
	Total allocated to project		\$1,150,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.